A Curvature Decomposition of the Explicit Formula

for the Riemann Zeta Function

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Research Note · May 2026 · v13

Abstract

We study the second logarithmic derivative of the completed Riemann zeta function in the form

$$H_\xi(\sigma,t) \;:=\; \partial_\sigma^2 \log|\xi(\sigma+it)|^2,$$

and interpret it as a second-derivative specialisation of the classical explicit formula. This yields a prime-cutoff decomposition $H_\xi = H_{\mathrm{local}} + H_{\mathrm{dual}}$ into local curvature contributions from the archimedean place and the finite Euler factors, together with a bookkeeping remainder H_{dual} defined as the finite-cutoff difference $H_\xi - H_{\mathrm{local}}$. The note also discusses the automorphic Rankin–Selberg analogue and a structural comparison of the curvature field with Li-type positivity criteria, noting that local positivity does not imply global Li positivity.

What is proved. Local positivity at every finite place: $V_p(\sigma) \geq 0$ for all primes p and all $\sigma > 0$ (Lemma 3.1). Together with the classical archimedean log-convexity $\psi_1(\sigma) \geq 0$, this yields non-negativity of the truncated local curvature $H_{\mathrm{local}}(\sigma,\kappa) \geq 0$ (Corollary 3.2). Critical divergence on the line $\sigma = \frac{1}{2}$: $H_{\mathrm{local}}(\frac{1}{2},\kappa) \asymp C(\log\kappa)^2$ as $\kappa \to \infty$, with constant $C > 0$ depending on the local-norm normalisation (Lemma 4.1). Subcritical convergence: $H_{\mathrm{local}}(\sigma,\kappa)$ tends to a finite limit as $\kappa \to \infty$ for every $\sigma > \frac{1}{2}$ (Lemma 4.2).

What is numerical. At a representative off-critical point $(\sigma,t) = (0.3, 0)$ with $\kappa = 1300$, $H_{\mathrm{local}} \approx 915$ while $H_\xi \approx 0.09$, so the bookkeeping remainder H_{dual} cancels the local curvature to better than 99.99% (Observation 5.1). Direct evaluation at the rounded ordinate $t = 14.13$, at distance approximately 5×10^{-3} from the first non-trivial zero $\gamma_1 \approx 14.1347$, gives $H_\xi(\frac{1}{2},t) \approx +89\,577.92$, illustrating the singular kernel $(s-\rho)^{-2}$ that dominates near zeros (Observation 5.2).

What remains open. Whether the singular test function implicit in H_ξ — whose Mellin side produces the kernel $(s-\rho)^{-2}$ — can be embedded into, or approximated by, the class of admissible test functions for which Weil positivity is equivalent to the Riemann Hypothesis (Open Problem 8.1).

Structure. The formal core in §§ 2–4 is non-circular and proof-complete. No use of the Riemann Hypothesis, the GUE conjecture, or the Hilbert–Pólya postulate is made anywhere in §§ 2–4.

Notation and Conventions

Critical line. Throughout, $s = \sigma + it$ with $\sigma = \Re(s)$ and $t = \Im(s)$; the critical line means $\sigma = \frac{1}{2}$. The completed zeta function is $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, so that $\xi(s) = \xi(1-s)$ on the whole plane.

Global parameters. The variables of this paper are $\sigma > 0$ (a real ordinate of evaluation), $t \in \mathbb{R}$ (a real imaginary part), and the prime cutoff $\kappa \geq 2$ (varied across the divergence and convergence statements). The cutoff κ runs over integers when summing $\sum_{p \leq \kappa}$ with p prime; equivalently, κ indexes the prime $p_{\pi(\kappa)}$. No further parameters enter the formal core.

Main objects. The curvature field is

$$H_\xi(\sigma, t) \ := \ \partial_\sigma^2 \log|\xi(\sigma + it)|^2,$$

and the truncated local curvature at cutoff κ is

$$H_{\text{local}}(\sigma, \kappa) \ := \ \psi_1(\sigma) \ + \ \sum_{p \leq \kappa} V_p(\sigma),$$

with archimedean contribution $\psi_1(\sigma) = \frac{1}{8}\psi^{(1)}(\sigma/2)$ ($\psi^{(1)}$ the trigamma function) and local Euler-factor contribution $V_p(\sigma) = 4(\log p)^2 p^{-2\sigma}/(1 - p^{-2\sigma})^2$. The bookkeeping remainder is $H_{\text{dual}}(\sigma, t, \kappa) := H_\xi(\sigma, t) - H_{\text{local}}(\sigma, \kappa)$. The non-trivial zeros of $\zeta(s)$ are denoted ρ; the first few imaginary ordinates are $\gamma_1 \approx 14.135$, $\gamma_2 \approx 21.022$.

Not to be confused with. The truncated local curvature $H_{\text{local}}(\sigma, \kappa)$ used in this paper is the prime-side sum of squared local Euler-factor contributions associated with the second-derivative specialisation of the explicit formula; it is not the classical von Mangoldt prime sum $\sum_{p^k \leq x} \Lambda(p^k)$ and not the renormalised prime weight that appears in admissible-test-function constructions for the Weil functional. The local positivity $V_p(\sigma) \geq 0$ established below is a place-by-place positivity statement and does *not* imply Li positivity of the coefficients λ_n (§ 7).

1 Introduction

1.1 Background and Motivation

This is the first paper in a series developing an explicit, finite-cutoff framework for studying the distribution of the non-trivial zeros of the Riemann zeta function $\zeta(s)$. The organising idea of the series is that prime-side quantities at finite cutoff carry arithmetic information about the zero geometry that is accessible by classical analytic means. The present paper introduces the central object of the series, the prime-cutoff decomposition of the second logarithmic derivative of the completed zeta function ξ.

The Hadamard product representation of the entire function ξ, together with logarithmic differentiation, gives

$$(\log \xi)''(s) \;=\; -\sum_{\rho} \frac{1}{(s-\rho)^2}, \tag{1}$$

where the sum runs over the non-trivial zeros ρ of ζ. The real part of (1), evaluated as a function of σ at fixed t, defines the curvature field H_ξ. On the local (prime and archimedean) side, the present paper introduces a separately constructed quantity, the truncated local curvature $H_{\mathrm{local}}(\sigma, \kappa)$, built in the Tate framework from per-place log-Mellin-norm contributions: an archimedean trigamma term $\psi_1(\sigma)$ and a finite Euler-factor sum $\sum_{p\le\kappa} V_p(\sigma)$ with closed forms specified in § 2. The bookkeeping remainder is defined by $H_{\mathrm{dual}}(\sigma, t, \kappa) = H_\xi(\sigma, t) - H_{\mathrm{local}}(\sigma, \kappa)$; in this paper it serves only as a bookkeeping device for the decomposition, and no spectral interpretation is imposed on it.

The decomposition $H_\xi = H_{\mathrm{local}} + H_{\mathrm{dual}}$ is not a new identity but a reorganisation of the explicit formula in curvature form, specialised to the kernel $2\Re(s-\rho)^{-2}$. The classical reference for the explicit formula and its specialisations is Edwards [11]. Burnol's invariant-analysis approach to the explicit formula [7] treats the local terms uniformly across all places, and the Connes–Consani trace-formula work at the archimedean place [2] establishes positivity statements there for compactly supported test functions. Neither of these isolates the prime-cutoff second-derivative decomposition $H_\xi = H_{\mathrm{local}} + H_{\mathrm{dual}}$ used here, in which, at every finite cutoff κ, the truncated local curvature $H_{\mathrm{local}}(\sigma, \kappa)$ is the sum of the archimedean trigamma term $\psi_1(\sigma)$ and the explicit closed-form Euler-factor contributions $V_p(\sigma)$. Equivalently, the decomposition $H_\xi = H_{\mathrm{local}} + H_{\mathrm{dual}}$ is the classical explicit formula [11] truncated to primes $p \le \kappa$ and applied *formally* — at the level of singular test functions and distributions — to the test function whose Mellin side produces the kernel $(s-\rho)^{-2}$. Rigorous embedding of this singular test function into the admissible class S_{ad} of the Weil functional is the subject of Open Problem 8.1. At the level of the formal identity, H_{local} is the truncated prime-side and H_{dual} is the remaining zero-side.

The present construction, whose finite-place positivity is the main result of this note, should be distinguished from established operator-theoretic programmes for the Riemann Hypothesis: it is not a Hilbert–Pólya spectral realisation in the sense of Connes [1], nor a Berry–Keating type quantum-Hamiltonian construction, nor a de Branges structure-function model. The vocabulary of "curvature" used here refers solely to the second-derivative origin of $V_p(\sigma)$ in $\log|\xi(\sigma+it)|^2$; no further geometric content is asserted. To the best of our knowledge, the prime-cutoff curvature formulation $H_\xi = H_{\mathrm{local}} + H_{\mathrm{dual}}$ does not appear explicitly in the literature (cf. J.-F. Burnol, personal communication).

1.2 Main Results

The four formal statements of this paper concern the prime-side object $H_{\mathrm{local}}(\sigma, \kappa)$ alone. First, $V_p(\sigma) \geq 0$ at every prime p and every $\sigma > 0$ (§ 3, Lemma 3.1), hence $H_{\mathrm{local}}(\sigma, \kappa) \geq 0$ throughout (Corollary 3.2). Second, on the critical line, the truncated local curvature diverges with order $(\log \kappa)^2$ as $\kappa \to \infty$ (§ 4, Lemma 4.1). Third, above the critical line the truncated local curvature tends to a finite limit (Lemma 4.2). The critical line is therefore the phase boundary between divergence and convergence of the local curvature sum.

The numerical illustrations of § 5 quantify the cancellation between H_{local} and H_{dual} off the critical line, and the singular peak of H_ξ near non-trivial zeros.

The Rankin–Selberg discussion of § 6 and the Li-coefficient comparison of § 7 are informational only. Their role is to place the curvature decomposition within the classical analytic-number-theory landscape and to delimit explicitly what local positivity does *not* imply.

The single open problem (§ 8, Open Problem 8.1) asks whether the singular test function implicit in H_ξ can be embedded into the class of admissible test functions for which Weil positivity is equivalent to the Riemann Hypothesis.

1.3 Reader Contract

This paper *proves* four statements about the truncated local curvature: place-by-place positivity at finite primes (Lemma 3.1); non-negativity of the truncated sum $H_{\mathrm{local}}(\sigma, \kappa)$ (Corollary 3.2); critical divergence of order $(\log \kappa)^2$ on $\sigma = \frac{1}{2}$ (Lemma 4.1); and convergence to a finite limit for $\sigma > \frac{1}{2}$ (Lemma 4.2). These are unconditional mathematical statements.

This paper *establishes numerically* two finite computations from the stated parameters (Observations 5.1–5.2): the off-critical compensation between H_{local} and H_{dual} at $(\sigma, t, \kappa) = (0.3, 0, 1300)$, and the singular peak of H_ξ near γ_1 on the critical line. Both numerical statements are finite computations from the stated parameters and are labelled [numerical] in their theorem headings.

This paper *leaves open* the embedding of the singular test function implicit in H_ξ into the admissible test-function class of the Weil functional (Open Problem 8.1, § 8). A positive answer would connect the curvature decomposition to the Weil positivity framework; whether such a connection extends to a full Weil positivity statement on the relevant admissible class is a separate and deeper question.

2 The Curvature Field

The Hadamard product representation of the entire function $\xi(s)$, together with logarithmic differentiation, yields

$$H_\xi(\sigma, t) \;:=\; \frac{d^2}{d\sigma^2}\log|\xi(\sigma+it)|^2 \;=\; 2\,\Re(\log\xi)''(s) \;=\; -2\,\Re\sum_\rho \frac{1}{(s-\rho)^2}, \tag{2}$$

where the sum runs over the non-trivial zeros ρ of $\zeta(s)$ and $s = \sigma + it$. The quantity H_ξ is the spectral (zero) side of an explicit-formula identity at second-derivative level: choosing a test function whose Mellin transform produces the kernel $(s-\rho)^{-2}$ recovers H_ξ as the spectral output.

The local (prime and archimedean) side is constructed separately, in the adelic framework of Tate's thesis, as a sum of per-place log-Mellin-norm contributions. At the archimedean place,

$$\psi_1(\sigma) \;:=\; \tfrac{1}{4}\,\frac{d^2}{d\sigma^2}\log|\Gamma(\sigma/2)|^2. \tag{3}$$

For real σ this coincides with $\frac{1}{8}\psi^{(1)}(\sigma/2)$, where $\psi^{(1)}$ denotes the trigamma function; one has $\frac{d^2}{d\sigma^2}\log|\Gamma(\sigma/2)|^2 = \frac{1}{2}\psi^{(1)}(\sigma/2)$ by the chain rule, so $\psi_1(\sigma) = \frac{1}{4}\cdot\frac{1}{2}\psi^{(1)}(\sigma/2) = \frac{1}{8}\psi^{(1)}(\sigma/2)$. At each finite place p, with M_s the local Mellin transform and f_p the standard Tate test function,

$$V_p(\sigma) \;:=\; \frac{d^2}{d\sigma^2}\log\|M_s f_p\|^2_{L^2(\mathbb{Q}_p^\times)}. \tag{4}$$

Explicitly, for the unramified local factor, $\|M_s f_p\|^2 = (1-p^{-1})(1-p^{-2\sigma})^{-1}$, so that

$$V_p(\sigma) \;=\; 4(\log p)^2\,\frac{p^{-2\sigma}}{(1-p^{-2\sigma})^2}. \tag{5}$$

The truncated local curvature at cutoff κ is

$$H_{\text{local}}(\sigma,\kappa) \;:=\; \psi_1(\sigma) \;+\; \sum_{p\le\kappa} V_p(\sigma), \tag{6}$$

and the bookkeeping remainder is

$$H_{\text{dual}}(\sigma,t,\kappa) \;:=\; H_\xi(\sigma,t) \;-\; H_{\text{local}}(\sigma,\kappa). \tag{7}$$

For $\sigma > \frac{1}{2}$, $H_{\text{local}}(\sigma,\kappa)$ converges as $\kappa\to\infty$ (Lemma 4.2) and $H_{\text{dual}}(\sigma,t,\kappa)$ has a well-defined limit in κ. At $\sigma = \frac{1}{2}$, H_{local} diverges (Lemma 4.1); the decomposition $H_\xi = H_{\text{local}} + H_{\text{dual}}$ is then interpreted at every finite κ, and H_{dual} is treated as a bookkeeping object without spectral interpretation.

The decomposition $H_\xi = H_{\text{local}} + H_{\text{dual}}$ holds at every finite cutoff κ by the definition of H_{dual} in (7). Read as a formal identity, it is the second-derivative specialisation of the truncated explicit formula to the singular test function whose Mellin side produces $(s-\rho)^{-2}$; rigorous treatment of that test function is the subject of Open Problem 8.1. It is not a new identity but a reorganisation in this formal sense, holding at every finite cutoff κ.

3 Local Positivity

Lemma 3.1 (Local positivity at finite places; proved)**.** *For every prime p and every $\sigma > 0$,*

$$V_p(\sigma) \;\geq\; 0.$$

Proof. From the explicit expression (5),

$$V_p(\sigma) \;=\; 4(\log p)^2\,\frac{p^{-2\sigma}}{(1-p^{-2\sigma})^2}.$$

For $\sigma > 0$ one has $0 < p^{-2\sigma} < 1$, hence the numerator and denominator are both strictly positive, so $V_p(\sigma) > 0$.

The same conclusion follows from a more general expression that is useful as a structural identity. With the active local norm at prime cutoff κ and annulus values $S_m = p^{1-m} - p^{-\kappa}$ for $m = 1, \ldots, \kappa$ and $S_0 = 2 - p^{-1} - p^{-\kappa}$, the Mellin norm of the level-κ test function $f_{\kappa,p}$ admits the representation

$$\|M_s f_{\kappa,p}\|^2 \;=\; (1-p^{-1})\Big(\frac{|S_0|^2}{1-p^{-2\sigma}} + \sum_{m=1}^{\kappa} |S_m|^2\, p^{2m\sigma}\Big),$$

which is a sum of exponentials in σ with positive coefficients. The logarithm of a sum of positive exponentials is convex, since its second derivative coincides with the variance of the exponential weights against the corresponding probability distribution and is therefore non-negative (Cauchy–Schwarz). ☐

The archimedean contribution $\psi_1(\sigma) \geq 0$ is classical, following from the log-convexity of the Gamma function (Bohr–Mollerup theorem; cf. Artin [9]). Combining the two place-wise positivities yields the following.

Corollary 3.2 (Truncated local curvature is non-negative; proved)**.** *For every $\sigma > 0$ and every $\kappa \geq 2$,*

$$H_{\mathrm{local}}(\sigma,\kappa) \;\geq\; 0.$$

Place-by-place positivity does not by itself approach the Riemann Hypothesis (cf. § 7); its role in this paper is to make the prime-side mass of H_ξ explicit, so that the cancellation between H_{local} and the bookkeeping remainder H_{dual} becomes visible at every finite cutoff. The numerical observations of § 5 quantify this cancellation off the critical line, where it is near-total, and on the critical line, where H_ξ exhibits singular peaks near the non-trivial zeros.

4 Divergence and Convergence

The behaviour of the truncated local curvature as $\kappa \to \infty$ separates the half-plane into two regimes. On the critical line, H_{local} diverges with order $(\log \kappa)^2$; for every $\sigma > \frac{1}{2}$, H_{local} tends to a finite limit.

Lemma 4.1 (Critical divergence; proved)**.** *As $\kappa \to \infty$,*

$$H_{\mathrm{local}}(\tfrac{1}{2}, \kappa) \;\asymp\; C\,(\log \kappa)^2$$

for a constant $C > 0$ depending on the local-norm normalisation.

Proof. From (5), $V_p(\frac{1}{2}) = 4(\log p)^2 p^{-1}(1 - p^{-1})^{-2}$, so for large p,

$$V_p(\tfrac{1}{2}) \;=\; 4(\log p)^2\, p^{-1}\,(1 + o(1)).$$

Partial summation against the prime number theorem $\pi(x) \sim x/\log x$ yields the standard Mertens-type estimate

$$\sum_{p \le \kappa} \frac{(\log p)^2}{p} \;=\; \tfrac{1}{2}(\log \kappa)^2 \;+\; O(\log \kappa) \qquad (\kappa \to \infty)$$

(Montgomery–Vaughan [10], § 2.2), hence $\sum_{p \le \kappa} V_p(\frac{1}{2}) \asymp C(\log \kappa)^2$ for a constant $C > 0$ depending on the local-norm convention. The archimedean contribution $\psi_1(\frac{1}{2})$ is bounded, so $H_{\mathrm{local}}(\frac{1}{2}, \kappa) \asymp C(\log \kappa)^2$. The leading constant is conventional; numerically one observes $C \approx 2$ for the standard Tate test functions with V_p as in (5). The proof asserts only the order of growth. ☐

Lemma 4.2 (Subcritical convergence; proved)**.** *For every $\sigma > \frac{1}{2}$,*

$$H_{\mathrm{local}}(\sigma, \kappa) \;\to\; H_{\mathrm{local}}(\sigma, \infty) \;<\; \infty \qquad (\kappa \to \infty).$$

Proof. From (5), $V_p(\sigma) = O((\log p)^2 p^{-2\sigma})$. For $\sigma > \frac{1}{2}$ one has $2\sigma > 1$, hence the series $\sum_p (\log p)^2 p^{-2\sigma}$ converges absolutely, by comparison with $\sum_p p^{-2\sigma+\delta}$ for any $\delta \in (0, 2\sigma - 1)$. The archimedean contribution $\psi_1(\sigma)$ is finite for $\sigma > 0$. The result follows. ☐

The critical line is therefore the phase boundary between divergence and convergence of the local curvature sum.

5 Numerical Illustration

The two observations of this section quantify the cancellation between H_{local} and H_{dual} at a representative off-critical point and the singular peak of H_ξ near a non-trivial zero on the critical line.

Observation 5.1 (Off-critical compensation; numerical). *At* $(\sigma, t) = (0.3, 0)$ *with* $\kappa = 1300$*,*

$$H_{\text{local}} \;\approx\; 915, \qquad H_\xi \;\approx\; 0.09.$$

The bookkeeping remainder H_{dual} *therefore cancels* H_{local} *to better than* 99.99%*.*

Observation 5.2 (Sampled values near zeros; numerical). *Take the rounded ordinates* $t_1 = 14.13$*, at distance approximately* 5×10^{-3} *from the first non-trivial zero* $\gamma_1 \approx 14.1347$*, and* $t_2 = 21.02$*, at distance approximately* 2×10^{-3} *from the second non-trivial zero* $\gamma_2 \approx 21.0220$*. Direct evaluation of* $H_\xi(\sigma, t) := \partial_\sigma^2 \log|\xi(\sigma + it)|^2$ *at these ordinates gives*

$$H_\xi(0.3, t_1) \;\approx\; -49.78, \qquad H_\xi(\tfrac{1}{2}, t_1) \;\approx\; +89\,577.92,$$

and on the critical line near the second zero, $H_\xi(\frac{1}{2}, t_2) \approx +480\,754.91$*.*

The spike at $\sigma = \frac{1}{2}$ arises because the kernel $2\Re(s-\rho)^{-2}$ in (2) has a double pole at $s = \rho$; near a zero this term dominates the spectral side, producing the large values observed. The larger value at t_2 in Observation 5.2 reflects the smaller sampling distance $|t_2 - \gamma_2| \approx 2 \times 10^{-3}$ to γ_2, compared with $|t_1 - \gamma_1| \approx 5 \times 10^{-3}$ to γ_1, together with the $|s - \rho|^{-2}$ singularity of the kernel. The functional-equation symmetry $H_\xi(\sigma, t) = H_\xi(1 - \sigma, t)$, which follows from $\xi(s) = \xi(1 - s)$, holds to machine precision in all numerical evaluations performed.

Together, Observations 5.1 and 5.2 quantify the cancellation between the place-by-place non-negative H_{local} (Lemma 3.1, Corollary 3.2) and the bookkeeping remainder H_{dual}: numerically, the cancellation reaches better than 99.99% off the critical line, while on the critical line H_ξ takes large positive values when (σ, t) is sampled close to a non-trivial zero, consistent with the singular kernel $(s - \rho)^{-2}$. The relationship is observational at finite cutoff and is not asserted as a theorem.

6 Rankin–Selberg Universality

This section is informational. The statement below is not used in any proof of this paper, and the universality claim relies on standard inputs from Rankin–Selberg theory cited inline; no theorem is asserted.

The divergence at $\sigma = \frac{1}{2}$ established in Lemma 4.1 is not specific to $\zeta(s)$. For a cuspidal automorphic representation π on $\mathrm{GL}(n)/\mathbb{Q}$ with Satake parameters $\alpha_{\pi,j}(p)$, the diagonal curvature sum

$$\sum_{p \le \kappa} \Big(\sum_{j=1}^{n} |\alpha_{\pi,j}(p)|^2 \Big) \frac{(\log p)^2}{p}$$

grows of order $C_\pi (\log \kappa)^2$ with $C_\pi > 0$. This follows from the simple pole of the Rankin–Selberg L-function $L(s, \pi \times \widetilde{\pi})$ at $s = 1$, combined with a Tauberian argument; the precise asymptotic coefficient depends on normalisation conventions (cf. Kowalski–Michel [8] for related prime sums in the Rankin–Selberg context). The curvature decomposition and the critical divergence are therefore universal phenomena for automorphic L-functions with Euler product and functional equation, on the line where the Rankin–Selberg L-function has its simple pole.

7 Connection to Li Coefficients

The Li coefficients λ_n, defined via

$$g(z) := \log \xi\left(\frac{1}{1-z}\right), \qquad g'(z) = \sum_{n\geq 1} \lambda_n \, z^{n-1},$$

satisfy Li's criterion [6]:

$$\text{RH} \iff \lambda_n \geq 0 \quad \text{for all } n \geq 1.$$

Differentiating once more,

$$g''(z) = (\log \xi)''\left(\frac{1}{1-z}\right)(1-z)^{-4} + 2(\log \xi)'\left(\frac{1}{1-z}\right)(1-z)^{-3}, \tag{8}$$

so that the term $(\log \xi)''(s)$ appearing in (8) is exactly the core of the curvature field $H_\xi(s) = 2\Re(\log \xi)''(s)$. The present curvature quantities are therefore structurally close to Li-type positivity criteria: both are built from logarithmic derivatives of ξ. The difference is that H_ξ is a continuous pointwise second-derivative field, whereas the Li coefficients package the same logarithmic data into a discrete sequence after the Möbius reparametrisation around $s = 1$.

Remark. *The local positivity $V_p(\sigma) \geq 0$ at each finite place established in Lemma 3.1 does* not *imply $\lambda_n \geq 0$. Li positivity is a global statement mixing all places, the functional equation, the complete zero geometry, and a specific test-function structure on the disc parametrisation. Place-by-place positivity is a necessary ingredient but not sufficient for Li positivity, and the gap between the two is of Riemann-Hypothesis strength.*

8 Open Problems

The central open problem arising from the curvature decomposition of this paper is the following.

Open Problem 8.1 (Embedding into the Weil framework)**.** *Does there exist a global packaging — a summation procedure, test-function class, or regularisation — that transfers the local positivity*

$$V_p(\sigma) \geq 0 \ (\forall\, p \, \textit{prime}), \qquad \psi_1(\sigma) \geq 0,$$

together with the critical divergence $H_{\text{local}}(\frac{1}{2}, \kappa) \to \infty$, into a global Li- or Weil-type positivity statement?

More precisely: can the singular test function implicit in H_ξ — whose Mellin side produces the kernel $(s-\rho)^{-2}$ — be embedded in, or approximated by, the class S_{ad} of admissible test functions for which Weil positivity $W(f * \check{f}) \geq 0$ is equivalent to the Riemann Hypothesis?

A positive answer to Open Problem 8.1 would connect the curvature decomposition to the Weil positivity framework (Weil [3]; Lagarias [4]; Connes–Consani [2]). Achieving full Weil positivity for the resulting admissible test function on the class S_{ad} would, by Weil's criterion (Weil [3]; Bombieri–Lagarias [5]), be equivalent to the Riemann Hypothesis itself. The present Open Problem is therefore the prior step of fitting $H_{\mathrm{local}}(\sigma, \kappa)$ into the admissible framework, and is not the positivity step itself; the latter remains a separate and deeper open question, and is not addressed here.

9 Non-Circularity

We document explicitly that the results of this paper do not rely on the objects they are directed toward.

No RH as input. The Riemann Hypothesis is not assumed at any point. The non-trivial zeros ρ enter the curvature field H_ξ through the Hadamard product representation of ξ, which is unconditional; no property of the location of the zeros is used in the formal core §§ 2–4.

No GUE, Montgomery, or Hilbert–Pólya hypothesis. No correlation statistic of the zeros and no postulated spectral realisation of the zeros is assumed or used. The terminology of "curvature" refers solely to the second-derivative origin of $V_p(\sigma)$ in $\log|\xi(\sigma + it)|^2$ and is not used as a geometric or operator-theoretic input.

No Li positivity as input. The Li criterion of § 7 is cited only as a comparison object; positivity of the coefficients λ_n is not assumed and not used in any proof of this paper. The Remark in § 7 explicitly states that local positivity does not imply Li positivity.

No Weil positivity as input. The Weil functional $W(f * \check{f})$ and the admissible class S_{ad} enter only in the formulation of Open Problem 8.1. No positivity statement on W is assumed in any of Lemmas 3.1, 4.1, 4.2, or in Corollary 3.2.

Inputs from classical analytic number theory. The formal core of the paper uses only standard inputs: the Hadamard product for ξ, log-convexity of the Gamma function (Bohr–Mollerup), the prime number theorem $\pi(x) \sim x/\log x$, and elementary partial summation. The Rankin–Selberg discussion in § 6 relies additionally on the simple pole of $L(s, \pi \times \widetilde{\pi})$ at $s = 1$ and a Tauberian argument; this section is informational and not used in the formal core.

Acknowledgments

The author thanks Jean-François Burnol for confirming that the prime-cutoff curvature formulation does not appear in the literature known to him, and for suggesting connections to the noncommutative-geometry programme of Connes and to the Langlands framework. The author also thanks the readers who provided feedback on earlier versions of this note.

Call for Feedback

This paper is part of an ongoing open research initiative toward the Riemann Hypothesis. All papers in the series are freely available on Zenodo under CC BY 4.0, and the accompanying verification code is hosted on GitHub.

Zenodo (`https://zenodo.org/search?q=Ulrich+Tehrani`) and GitHub (`https://github.com/utehrani/analysislab-nt`).

Topics of particular interest for diagnostic feedback on this note include: (i) whether the singular kernel $(s-\rho)^{-2}$ underlying H_ξ admits a natural regularisation or limit construction in S_{ad}, in the sense of Open Problem 8.1; and (ii) whether the Rankin–Selberg analogue (§ 6) admits a refined asymptotic with explicit leading constant under standard conventions.

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